



GenAI SECURITY
PROJECT
TOP 10 FOR LLM AND GENERATIVE AI

AI Red Teaming Solutions Taxonomy

GenAI Solutions Landscape Initiative

Version 1.0
March 4, 2026

Table of Content

Introduction	2
Purpose of This Taxonomy	2
Capability Categories	2
Red Teaming (Offense / Attack Simulation)	3
Blue Teaming (Defense / Detection / Response)	3
Purple Teaming (Continuous Attack-Defense Fusion)	3
Shared Capabilities	3
Capability Matrix	3
Lifecycle Coverage	4
Intended Audience	4
Capabilities Detail	5
1. Red Teaming Capabilities	5
Scope / Plan	5
Data Augmentation & Fine-Tuning	5
Development & Experimentation	5
Test & Evaluation	6
Release	6
Deploy	6
Operate	6
Monitor	6
Govern	7
2. Blue Teaming Capabilities	7
Scope / Plan	7
Data Augmentation & Fine-Tuning	7
Development & Experimentation	7
Test & Evaluation	7
Release	8
Deploy	8

Operate	8
Monitor	8
Govern	8
3. Purple Teaming Capabilities	9
Scope / Plan	9
Data Augmentation & Fine-Tuning	9
Development & Experimentation	9
Test & Evaluation	9
Release	10
Deploy	10
Operate	10
Monitor	10
Govern	11
4. Shared Capabilities	11
Scope / Plan	11
Data Augmentation & Fine-Tuning	11
Development & Experimentation	11
Release & Operate	11
Govern	11

Introduction

Gen AI & Agentic Red Teaming Solutions Guide Taxonomy

As organizations increasingly deploy generative AI and autonomous agents into business-critical workflows, traditional application security practices are no longer sufficient. AI systems introduce new classes of risk including prompt injection, model misuse, agent privilege escalation, data poisoning, hallucinations, and emergent behaviors that evolve continuously throughout the AI adoption lifecycle.

Gen AI and Agentic Red Teaming provides a structured, lifecycle-wide approach to identifying, measuring, mitigating, and governing these risks through coordinated adversarial testing, defensive validation, and continuous feedback loops.

Purpose of This Taxonomy

This solutions guide defines a capability taxonomy for AI and Agentic Red Teaming software, organized to help organizations:

- Evaluate and compare open source and commercial solutions in a rapidly evolving market
- Establish consistent AI security and risk management practices
- Align technical controls with governance, compliance, and executive oversight
- Integrate AI risk management into existing DevSecOps and SecOps workflows

The taxonomy is grounded in the Gen AI and Agentic application and deployment lifecycle, recognizing that AI risk emerges not only at deployment, but from planning and development through operation and governance.

This document serves as the foundational taxonomy for the AI and Agentic Red Teaming Solutions Guide. It establishes a common language and structured classification of capabilities across red, blue, and purple teaming, organized by practitioner lifecycle stages. The taxonomy is intended to support companion solutions guides and cheat sheets that map, in detail, how specific tools, platforms, and solution categories cover each capability. By separating what capabilities are required from how solutions implement them, the taxonomy enables consistent comparison, clearer gap analysis, and repeatable evaluation of AI and agentic red teaming solutions as threats and technologies evolve.

Capability Categories

The framework organizes capabilities into four categories aligned to security teaming and collaborative security team roles.

Red Teaming (Offense / Attack Simulation)

Focuses on proactively discovering weaknesses in AI models, agents, data pipelines, and integrations by simulating malicious users, compromised agents, and supply-chain attacks.

Blue Teaming (Defense / Detection / Response)

Ensures that guardrails, policies, monitoring, and runtime protections prevent, detect, and respond to AI-specific threats across environments.

Purple Teaming (Continuous Attack–Defense Fusion)

Integrates red and blue activities into closed-loop feedback systems, ensuring adversarial findings directly improve defensive controls, policies, and operational posture.

Shared Capabilities

Foundational capabilities—such as asset inventory, telemetry, provenance, metrics, and audit artifacts—that enable consistency, automation, and governance across all teams.

Capability Matrix

OWASP GenAIOps/SecOps Lifecycle Stage	Red-Team Categories (offence / attack simulation)	Blue-Team Categories (defence / detection / response)	Purple-Team Categories (continuous attack-defence fusion)	Shared Capabilities
Scope / Plan	- Threat-model design aids - LLM/agent attack-surface mapping	- AI asset inventory - AI posture dashboards - e.g. AI-SPM/AI-TRISM	- Risk-scoring boards - Import Red scenarios - Map to Blue controls	- Risk taxonomy import/export - Visual data-flow mapping - Export of tests as stories
Data Aug & Fine-Tune	- Data-poison fuzzing - Synthetic insert generation - Malicious model artifacts	- Data-lineage / provenance - DLP scanning	- Bias-toxicity co-auditing - Replay Red mutations through Blue filters	- Diff on corpus versions - Bias/POII scorecards - Signed data packages
Dev & Experimentation	- Model vulnerability scanner - Jaillbreak - Bias - RCE - Agent-logic corruption testers	- SAST/DAST/IAST Scanning - LLM plug-ins - tools - code - infrastructure	- Interactive sandbox - Defender signal analysis	- Reasoning-trace capture - Auto-ticket for failed test - IDE plug-in
Test & Evaluation	- Automated adversarial suite - Prompt-chaining - Multi-turn - Protocol support A2A, MCP - RAG-poison scenario runners	- Guard-rail conformance - Policy testing/validation	- One-click "purple run" - Replays every Red case - Metrics exporting e.g. Blue metrics	- Success-threshold analysis - Hallucination vs mis-alignment labelling - CI hooks
Release	- AI-BOM - Supply-chain attestation checkers	- Secure CI/CD gates - Signing & provenance validation	Deployment "purple pipeline" analysis	- SBOM-diff on model & code - Release-risk dashboard - Rollback script generation
Deploy	- Tool-chain/plug-in misuse simulation - Agent privilege-escalation emulation - Test data isolation - cross-tenant	- LLM/Agent firewall - Policy Management	Live traffic chaos simulation	- Real-time policy shadow-mode - MCP/A2A channel spoofing - Cost-impact tracking
Operate	- Autonomous red bots - Continuous prompt fuzzing - Memory poisoning	- Runtime AI-SPM/AI-WAF - User and Entity Behavior Analytics (UEBA) for AI signals	- Closed-loop purple coach - Correlate red attacks with blue alerts - Rule tuning	- Agent-behaviour baselining - Trust-boundary alerting - Auto-guard-rail patch
Monitor	Synthetic-user & rogue-agent generation	- Posture & metric collection - User and Entity Behavior Analytics (UEBA) for AI signals	- Purple SIEM lens - Merged telemetry analysis/reporting - Red with Blue KPIs	- Time-series scoring - Adaptive hunt packs - Model-drift vs threat-drift diff analysis - Aggregate findings into risk-trend dashboards
Govern	Audit-grade attack-path replay & evidencing	Policy / compliance orchestration (AI-TRISM layer)	- Risk simulators - Residual risk analysis red/blue cycles	- Signed artifact store - Framework mapping (e.g. NIST AI RMF, OWASP, MITRE & ISO 5339, etc) - Executive reporting - Feed discoveries into retraining data and IR playbook

Lifecycle Coverage

Capabilities are mapped across the full AI lifecycle:

- **Scope / Plan** – Threat modeling, attack-surface mapping, asset inventory, and risk prioritization
- **Data Augmentation & Fine-Tuning** – Data integrity, bias, poisoning resilience, and provenance
- **Development & Experimentation** – Model and agent testing, tooling security, and developer workflows
- **Test & Evaluation** – Automated adversarial testing, guardrail validation, and policy enforcement
- **Release** – Supply-chain integrity, provenance, and risk acceptance gates
- **Deploy** – Runtime controls, policy enforcement, and environment isolation
- **Operate** – Continuous adversarial activity, detection, and response
- **Monitor** – Telemetry, drift analysis, and threat hunting
- **Govern** – Auditability, compliance alignment, executive reporting, and continuous improvement

This lifecycle-based structure ensures AI risk is addressed before, during, and after deployment, rather than treated as a point-in-time assessment.

Intended Audience

This guide is designed for:

- Security leaders preparing for AI-driven threat evolution
- AI platform and infrastructure owners
- Incident response and threat-hunting teams
- Risk and governance leaders overseeing AI adoption

As attackers increasingly weaponize AI, defenders must do the same, responsibly. AI-accelerated and Agentic Red Teaming provides the means to continuously simulate advanced adversaries, prepare infrastructure for AI-native threats, and harden systems against the growing scale, speed, and sophistication of AI-enabled attacks.

Capabilities Detail

Capabilities are mapped across the full AI lifecycle:

1. Red Teaming Capabilities

(Offense / Adversarial Simulation)

Scope / Plan

- **Threat-model design aids;** Structured tools that guide practitioners in identifying AI-specific threat actors, misuse scenarios, attack paths, and failure modes across models, agents, data pipelines, and integrations.
- **LLM / agent attack-surface mapping;** Capabilities that enumerate and visualize exposed prompts, tools, APIs, memory stores, plugins, and decision boundaries where adversarial interaction or exploitation may occur.

Data Augmentation & Fine-Tuning

- **Data-poison fuzzing;** Techniques that intentionally inject malformed, adversarial, or misleading training data to evaluate model sensitivity, robustness, and susceptibility to poisoning attacks.
- **Synthetic insert generation;** Generation of artificial malicious, biased, or edge-case data designed to stress test training and fine-tuning pipelines without using real-world sensitive data.
- **Malicious model artifacts;** Simulated tampered or compromised model files, weights, or metadata used to assess detection, validation, and integrity controls within AI supply chains.

Development & Experimentation

- **Model vulnerability scanning;** Automated testing of models for known and emerging weaknesses such as prompt injection, jailbreaks, bias amplification, unsafe completions, and unintended capability exposure.
- **Agent-logic corruption testing;** Evaluation of agent workflows under adversarial manipulation of reasoning chains, memory, tool outputs, or control logic to uncover unsafe autonomous behaviors.

Test & Evaluation

- **Automated adversarial suites;** Repeatable collections of adversarial test cases that exercise models and agents across common and novel attack patterns to establish baseline and comparative risk metrics.
- **Prompt-chaining attacks;** Testing techniques that exploit multi-step prompt dependencies to induce policy bypass, goal hijacking, or unintended behaviors not visible in single-prompt tests.
- **Multi-turn attacks;** Adversarial testing across extended interactions to identify emergent failures that arise only through sustained dialogue or state accumulation.
- **Protocol attacks (A2A, MCP);** Simulation of malicious use or spoofing of agent-to-agent and model communication protocols to test trust boundaries and authentication mechanisms.
- **RAG-poison scenario runners;** Execution of adversarial scenarios targeting retrieval-augmented generation pipelines by manipulating indexed data or retrieval logic to influence outputs.

Release

- **Supply-chain attack simulation;** Adversarial testing of model, data, and dependency delivery processes to assess exposure to compromised components, unauthorized changes, or provenance failures.

Deploy

- **Tool-chain / plug-in misuse simulation;** Testing of deployed tools and plugins to identify abuse paths where integrations enable privilege escalation, data exfiltration, or unintended side effects.
- **Agent privilege-escalation emulation;** Simulation of scenarios in which agents attempt to exceed intended authority, access restricted tools, or bypass control boundaries in deployed environments.
- **Cross-tenant data exposure testing;** Validation that deployed AI systems enforce tenant isolation and prevent unintended data leakage across users or organizational boundaries.

Operate

- **Autonomous red bots;** Self-directed adversarial agents that continuously probe deployed AI systems for weaknesses without requiring manual test execution.
- **Continuous prompt fuzzing;** Ongoing generation and mutation of prompts to discover new attack patterns and emergent failure modes in production or production-like systems.
- **Memory poisoning;** Testing of agent memory stores and long-term state mechanisms to evaluate susceptibility to persistent manipulation or corruption.

Monitor

- **Synthetic user & rogue-agent generation;** Creation of simulated malicious users or compromised agents to test detection and monitoring capabilities under realistic operating conditions.

Govern

- **Audit-grade attack-path replay;** Capabilities to reproduce adversarial tests and attacks with full context and evidence to support audits, compliance reviews, and post-incident analysis.

2. Blue Teaming Capabilities

(Defense / Detection / Response)

Scope / Plan

- **AI asset inventory;** Centralized and continuously updated catalog of AI models, agents, datasets, tools, plugins, and environments used to establish ownership and accountability.
- **AI posture dashboards (AI-SPM / AI-TRISM);** Dashboards that aggregate AI risk indicators, control coverage, and compliance signals to provide situational awareness across the AI estate.
- **Risk-scoring boards;** Mechanisms that quantify AI risks based on likelihood, impact, and control effectiveness to support prioritization and decision-making.

Data Augmentation & Fine-Tuning

- **Data lineage & provenance tracking;** Capabilities that trace the origin, transformation, and usage of training and fine-tuning data to support integrity, compliance, and incident response.
- **DLP scanning;** Detection of sensitive, personal, or regulated information within AI datasets to reduce privacy and regulatory risk.
- **Bias-toxicity co-auditing;** Joint assessment of fairness, bias, and harmful content characteristics in datasets and model outputs to prevent systemic harm.

Development & Experimentation

- **SAST / DAST / IAST scanning;** Application of traditional application security testing methods to AI-related code, services, and infrastructure components.
- **LLM plugin, tool, and infrastructure scanning;** Evaluation of third-party and internal AI integrations for vulnerabilities, misconfigurations, and unsafe behaviors.

Test & Evaluation

- **Guardrail conformance testing;** Validation that deployed guardrails consistently enforce content, safety, and behavioral policies under expected and adversarial conditions.
- **Policy testing & validation;** Systematic evaluation of policy definitions to ensure they are complete, correctly implemented, and aligned with organizational intent.

Release

- **Secure CI/CD gates;** Automated enforcement points that prevent AI artifacts from being released unless defined security and risk criteria are satisfied.
- **Signing & provenance validation;** Verification that released models, code, and data artifacts originate from trusted sources and have not been altered.

Deploy

- **LLM / agent firewall;** Runtime enforcement layer that filters inputs, outputs, and tool usage based on defined policies and risk thresholds.
- **Policy management;** Centralized systems for defining, versioning, and deploying AI security and safety policies across environments.

Operate

- **Runtime AI-SPM / AI-WAF;** Continuous monitoring and protection capabilities that detect and mitigate attacks against AI systems during operation.
- **Anomaly & drift detection;** Detection of deviations in model behavior, agent actions, or data distributions that may indicate failure or attack.
- **Trust-boundary alerting;** Alerting mechanisms that identify violations of defined trust boundaries between agents, systems, users, and data.

Monitor

- **Posture & metric collection;** Aggregation of operational, security, and risk metrics from deployed AI systems to support analysis and reporting.
- **UEBA for AI and agent signals;** Behavior-based analytics applied to AI and agent activity to detect abnormal or malicious patterns.

Govern

- **Policy & compliance orchestration (AI-TRiSM);** Capabilities that align AI operations with regulatory, legal, and internal governance requirements across the lifecycle.
- **Executive reporting;** High-level summaries of AI risk posture, trends, and residual risk designed for leadership and board audiences.

3. Purple Teaming Capabilities

(Integrated Red-Blue Feedback Loops)

Scope / Plan

- **Import red scenarios;** Ingestion of adversarial test cases and threat scenarios to enable coordinated defensive validation.
- **Map red scenarios to blue controls;** Traceability between attacks and defensive measures to identify gaps and measure coverage.

Data Augmentation & Fine-Tuning

- **Replay red mutations through blue filters;** Re-execution of known adversarial data patterns to confirm that defensive controls remain effective over time.
- **Corpus diffing;** Comparison of dataset versions to identify changes that may introduce new risks or regressions.

Development & Experimentation

- **Interactive sandbox;** Shared environment where red and blue teams collaboratively test attacks and defenses without production risk.
- **Defender signal analysis;** Analysis of detection outputs generated during adversarial testing to assess signal quality and coverage.
- **Reasoning-trace capture;** Collection of model or agent reasoning steps to support joint investigation and remediation.
- **Auto-ticketing for failed tests;** Automatic creation of remediation work items when adversarial tests reveal control failures.

Test & Evaluation

- **One-click purple runs;** Unified execution of adversarial tests with integrated defensive measurement and reporting.
- **Metrics exporting (blue KPIs);** Export of defensive metrics aligned to adversarial outcomes for continuous improvement.
- **Success-threshold analysis;** Definition and evaluation of acceptable performance and risk thresholds.
- **Hallucination vs misalignment labeling;** Classification of failures to distinguish factual errors from policy or alignment violations.
- **Continuous Integration hooks;** Integration of purple testing into continuous integration workflows.

Release

- **Purple pipeline analysis;** End-to-end assessment of release pipelines to identify residual risks across red and blue perspectives.
- **Release-risk dashboards;** Visualization of aggregated risks and mitigations at release time.
- **Rollback script generation;** Automated preparation of rollback procedures to reduce blast radius of failed releases.

Deploy

- **Live traffic chaos simulation;** Controlled disruption of live or mirrored traffic to test resilience and detection under stress.
- **Real-time policy shadow mode;** Observation of policy effects without enforcement to assess potential impact.
- **Protocol spoofing (MCP / A2A);** Testing of detection and response to spoofed or malicious inter-agent communications.
- **Cost-impact tracking;** Measurement of financial and operational costs associated with attacks and defenses.

Operate

- **Closed-loop purple coaching;** Automated feedback mechanisms that use attack results to continuously tune defenses.
- **Red/blue alert correlation;** Linking adversarial actions with defensive alerts to validate coverage and accuracy.
- **Rule tuning;** Refinement of defensive detection and enforcement rules based on observed failures.
- **Agent behavior baselining;** Establishment of normal behavior profiles to improve anomaly detection.
- **Auto guardrail patching;** Automated updates to guardrails in response to discovered weaknesses.

Monitor

- **Purple SIEM lens;** Unified analysis layer that combines adversarial and defensive telemetry.
- **Merged telemetry analysis;** Joint reporting that correlates red-team activity with blue-team performance.
- **Time-series scoring;** Tracking of risk and control effectiveness trends over time.
- **Adaptive hunt packs;** Dynamic threat-hunting strategies informed by recent adversarial findings.
- **Model-drift vs threat-drift analysis;** Differentiation between benign model evolution and adversarial manipulation.

Govern

- **Residual risk analysis cycles;** Ongoing assessment of remaining risks after mitigation efforts.
- **Risk simulators;** Forward-looking modeling of potential future AI risk scenarios.
- **Feedback to retraining & IR playbooks;** Systematic incorporation of adversarial findings into model improvement and incident response processes.

4. Shared Capabilities

(Cross-Category Foundations)

Scope / Plan

- **Risk taxonomy import/export;** Standardized representation of AI risks to enable interoperability across tools and frameworks.
- **Visual data-flow mapping;** Shared visualization of AI system dependencies and trust boundaries.
- **Export of tests as stories;** Human-readable representations of tests for communication and governance.

Data Augmentation & Fine-Tuning

- **Bias / PII scorecards;** Standardized reporting of sensitive data and bias indicators.
- **Signed data packages;** Cryptographically verifiable datasets used across teams.

Development & Experimentation

- **IDE plugins;** Developer-integrated tools that surface AI risks early in the lifecycle.

Release & Operate

- **AI-BOM / SBOM diffing;** Traceability and change detection across models, data, and code.

Govern

- **Framework mapping;** Alignment of capabilities to standards such as NIST AI RMF, OWASP, MITRE, and ISO.
- **Signed artifact stores;** Tamper-evident storage of evidence and artifacts for audit and compliance.